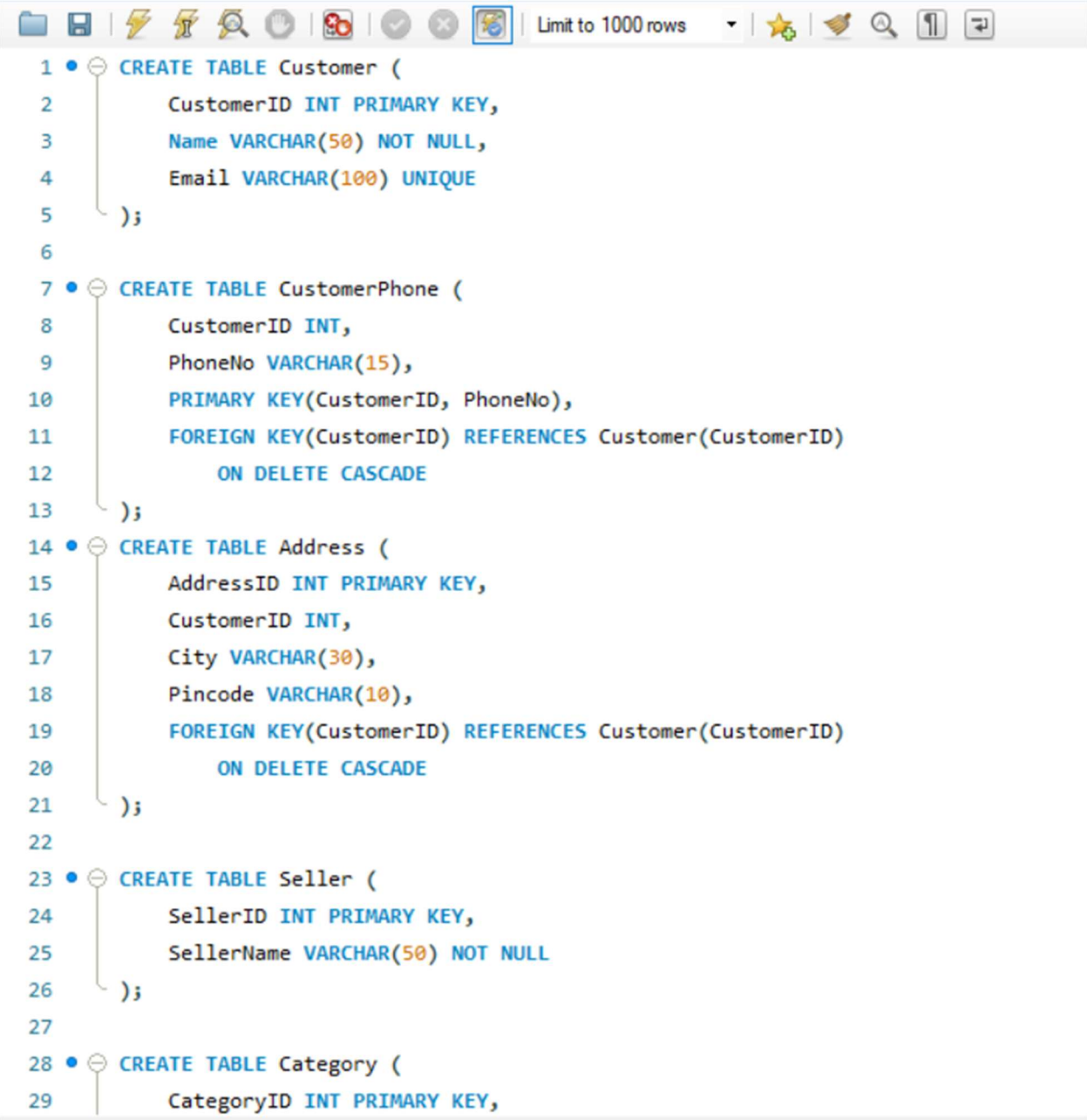
EXPERIMENT: 2

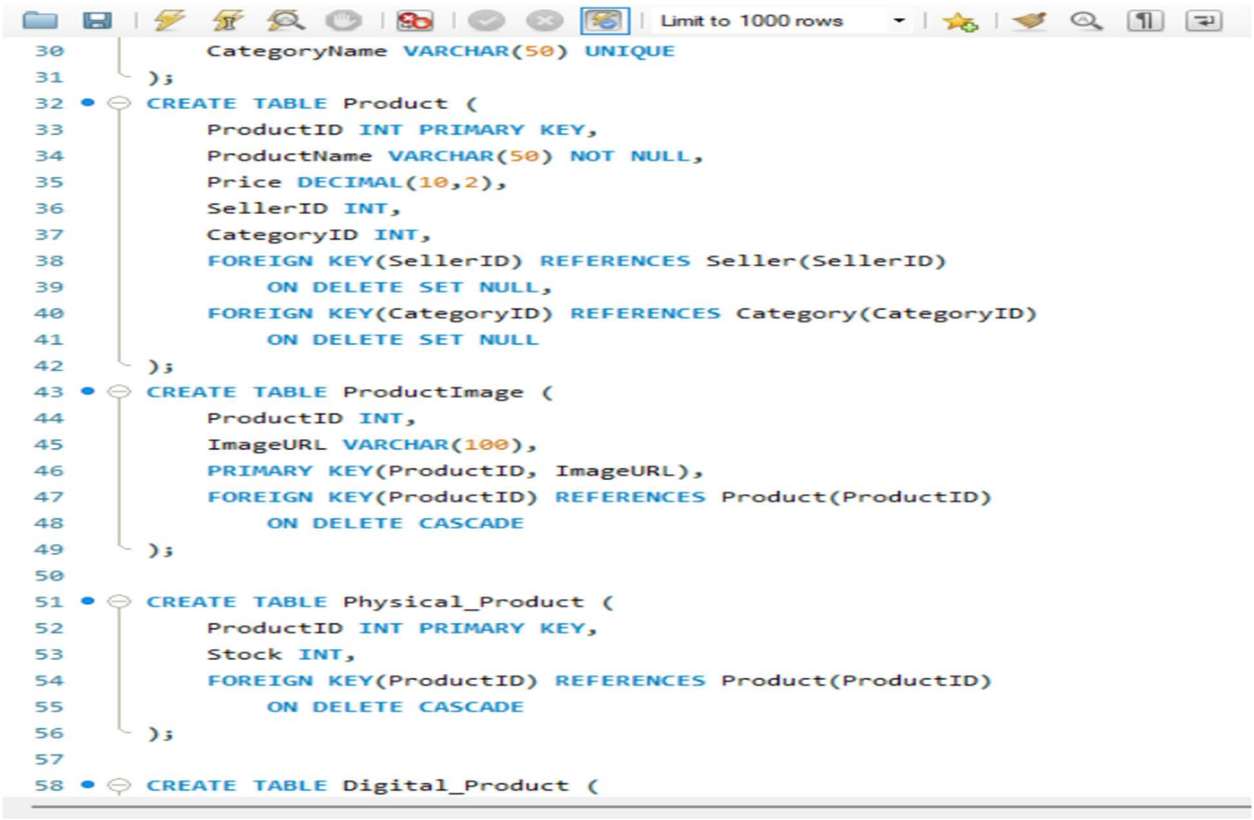
Aim:

To convert an ER diagram into a relational schema and implement it in MySQL using primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity.

Source Code:




```
1  CREATE TABLE Customer (
2      CustomerID INT PRIMARY KEY,
3      Name VARCHAR(50) NOT NULL,
4      Email VARCHAR(100) UNIQUE
5  );
6
7  CREATE TABLE CustomerPhone (
8      CustomerID INT,
9      PhoneNo VARCHAR(15),
10     PRIMARY KEY(CustomerID, PhoneNo),
11     FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
12         ON DELETE CASCADE
13 );
14 CREATE TABLE Address (
15     AddressID INT PRIMARY KEY,
16     CustomerID INT,
17     City VARCHAR(30),
18     Pincode VARCHAR(10),
19     FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
20         ON DELETE CASCADE
21 );
22
23 CREATE TABLE Seller (
24     SellerID INT PRIMARY KEY,
25     SellerName VARCHAR(50) NOT NULL
26 );
27
28 CREATE TABLE Category (
29     CategoryID INT PRIMARY KEY,
```



```
30      CategoryName VARCHAR(50) UNIQUE
31  );
32  CREATE TABLE Product (
33      ProductID INT PRIMARY KEY,
34      ProductName VARCHAR(50) NOT NULL,
35      Price DECIMAL(10,2),
36      SellerID INT,
37      CategoryID INT,
38      FOREIGN KEY(SellerID) REFERENCES Seller(SellerID)
39          ON DELETE SET NULL,
40      FOREIGN KEY(CategoryID) REFERENCES Category(CategoryID)
41          ON DELETE SET NULL
42  );
43  CREATE TABLE ProductImage (
44      ProductID INT,
45      ImageURL VARCHAR(100),
46      PRIMARY KEY(ProductID, ImageURL),
47      FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
48          ON DELETE CASCADE
49  );
50
51  CREATE TABLE Physical_Product (
52      ProductID INT PRIMARY KEY,
53      Stock INT,
54      FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
55          ON DELETE CASCADE
56  );
57
58  CREATE TABLE Digital_Product (
```

```
57
58  CREATE TABLE Digital_Product (
59      ProductID INT PRIMARY KEY,
60      FileType VARCHAR(20),
61      FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
62          ON DELETE CASCADE
63  );
64  CREATE TABLE Orders (
65      OrderID INT PRIMARY KEY,
66      CustomerID INT,
67      AddressID INT,
68      OrderDate DATE,
69      FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
70          ON DELETE CASCADE,
71      FOREIGN KEY(AddressID) REFERENCES Address(AddressID)
72          ON DELETE SET NULL
73  );
74
75  CREATE TABLE OrderItem (
76      OrderID INT,
77      ProductID INT,
78      Quantity INT,
79      PRIMARY KEY(OrderID, ProductID),
80      FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
81          ON DELETE CASCADE,
82      FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
83          ON DELETE CASCADE
84  );
85
```

```sql

CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY,
    OrderID INT UNIQUE,
    Amount DECIMAL(10,2),
    FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
        ON DELETE CASCADE
);
CREATE TABLE Delivery (
    DeliveryID INT PRIMARY KEY,
    OrderID INT UNIQUE,
    Status VARCHAR(20),
    FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
        ON DELETE CASCADE
);



-- Sample Data

INSERT INTO Customer VALUES
(1, 'Shreya', 'shreya@gmail.com');

INSERT INTO CustomerPhone VALUES
(1, '9876543210');

INSERT INTO Address VALUES
(101, 1, 'Lucknow', '226001');
```

```sql
INSERT INTO Address VALUES
(101, 1, 'Lucknow', '226001');

INSERT INTO Seller VALUES
(201, 'Tech World');

INSERT INTO Category VALUES
(301, 'Electronics');
INSERT INTO Product VALUES
(401, 'Laptop', 65000, 201, 301);

INSERT INTO Physical_Product VALUES
(401, 10);

INSERT INTO Orders VALUES
(501, 1, 101, '2026-08-19');

INSERT INTO OrderItem VALUES
(501, 401, 1);

INSERT INTO Payment VALUES
(601, 501, 65000);

INSERT INTO Delivery VALUES
(701, 501, 'Shipped');
INSERT INTO Orders VALUES
(502, 99, 101, '2026-08-19');
-- Display Output

SELECT * FROM Customer;
SELECT * FROM Product;
SELECT * FROM Orders;
```

OUTPUT:

Result Grid | Filter Rows:

CustomerID	Name	Email
1	Shreya	shreya@gmail.com
NULL	NULL	NULL

Result Grid | Filter Rows: | Edit:

ProductID	ProductName	Price	SellerID	CategoryID
401	Laptop	65000.00	201	301
NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows:

OrderID	CustomerID	AddressID	OrderDate
501	1	101	2026-08-19
NULL	NULL	NULL	NULL

Result Grid | Filter Rows:

OrderID	ProductID	Quantity
501	401	1
NULL	NULL	NULL

Result:

The given ER diagram was successfully converted into a relational schema and implemented in MySQL using Primary Key, Foreign Key, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints. Sample data was inserted and referential integrity was successfully demonstrated using a foreign key violation.